

Understanding Perplexity in Language Models

From Entropy to Cross-Entropy to Perplexity

Dr. Anish Roychowdhury, Plaksha University

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1 Introduction

This document provides a comprehensive understanding of perplexity in language models, starting from the fundamental concept of entropy and building up through cross-entropy to finally arrive at perplexity. We'll use concrete examples throughout to illustrate these concepts.

2 Information Theory: Entropy

2.1 The Concept of Entropy

Entropy, introduced by Claude Shannon, measures the average amount of information (or uncertainty) in a random variable. For a discrete random variable X with probability distribution $P(X = x_i) = p_i$, the entropy is defined as:

$$H(X) = - \sum_{i=1}^n p_i \log_2 p_i \quad (1)$$

where the logarithm is typically base 2 (measuring information in bits).

2.2 Intuition Behind Entropy

- Entropy measures the average "surprise" or uncertainty in outcomes
- Higher entropy = more uncertainty = more information needed to describe the system
- Lower entropy = more predictability = less information needed

2.3 Simple Example: Coin Flipping

Fair Coin: $P(H) = P(T) = 0.5$

$$H(X) = -[0.5 \log_2(0.5) + 0.5 \log_2(0.5)] \quad (2)$$

$$= -[0.5 \times (-1) + 0.5 \times (-1)] \quad (3)$$

$$= 1 \text{ bit} \quad (4)$$

Biased Coin: $P(H) = 0.9, P(T) = 0.1$

$$H(X) = -[0.9 \log_2(0.9) + 0.1 \log_2(0.1)] \quad (5)$$

$$= -[0.9 \times (-0.152) + 0.1 \times (-3.322)] \quad (6)$$

$$= 0.469 \text{ bits} \quad (7)$$

The biased coin has lower entropy because it's more predictable.

2.4 Vocabulary Example

Consider a simple vocabulary $V = \{\text{the, cat, sat, on, mat}\}$ with frequencies:

Word	Count	Probability
the	40	0.4
cat	20	0.2
sat	20	0.2
on	15	0.15
mat	5	0.05

The entropy is:

$$H = -[0.4 \log_2(0.4) + 0.2 \log_2(0.2) + 0.2 \log_2(0.2) + 0.15 \log_2(0.15) + 0.05 \log_2(0.05)] \quad (8)$$

$$= -[0.4(-1.32) + 0.2(-2.32) + 0.2(-2.32) + 0.15(-2.74) + 0.05(-4.32)] \quad (9)$$

$$= 2.04 \text{ bits} \quad (10)$$

3 Cross-Entropy

3.1 Definition

Cross-entropy measures the average number of bits needed to represent events from distribution P using a coding scheme optimized for distribution Q :

$$H(P, Q) = - \sum_{i=1}^n p_i \log q_i \quad (11)$$

where P is the true distribution and Q is the predicted/estimated distribution.

3.2 Key Properties

- $H(P, Q) \geq H(P)$ (cross-entropy is always at least as large as entropy)
- $H(P, Q) = H(P)$ if and only if $P = Q$
- Cross-entropy increases as Q deviates from P

3.3 Cross-Entropy in Classification

3.3.1 Binary Classification Example

Consider a binary classification problem where we want to classify emails as spam (1) or not spam (0).

True labels: $y_1 = 1, y_2 = 0, y_3 = 1, y_4 = 0$

Model predictions: $\hat{p}_1 = 0.9, \hat{p}_2 = 0.1, \hat{p}_3 = 0.7, \hat{p}_4 = 0.3$

The cross-entropy loss is:

$$CE = -\frac{1}{4} \sum_{i=1}^4 [y_i \log(\hat{p}_i) + (1 - y_i) \log(1 - \hat{p}_i)] \quad (12)$$

$$= -\frac{1}{4} [1 \cdot \log(0.9) + 0 \cdot \log(0.1) + 1 \cdot \log(0.7) + 0 \cdot \log(0.3)] \quad (13)$$

$$- \frac{1}{4} [0 \cdot \log(0.1) + 1 \cdot \log(0.9) + 0 \cdot \log(0.3) + 1 \cdot \log(0.7)] \quad (14)$$

$$= -\frac{1}{4} [\log(0.9) + \log(0.7) + \log(0.9) + \log(0.7)] \quad (15)$$

$$= -\frac{1}{4} [2 \log(0.9) + 2 \log(0.7)] \quad (16)$$

$$\approx 0.201 \quad (17)$$

3.3.2 Multi-class Classification

For a 3-class problem with true distribution $P = [0.7, 0.2, 0.1]$ and predicted distribution $Q = [0.6, 0.3, 0.1]$:

$$CE = -[0.7 \log(0.6) + 0.2 \log(0.3) + 0.1 \log(0.1)] \quad (18)$$

$$= -[0.7(-0.51) + 0.2(-1.20) + 0.1(-2.30)] \quad (19)$$

$$= 0.357 + 0.240 + 0.230 = 0.827 \quad (20)$$

4 Perplexity in Language Models

4.1 Definition

Perplexity is defined as the exponential of cross-entropy:

$$\text{Perplexity} = 2^{H(P,Q)} = \exp(CE) \quad (21)$$

For a language model predicting a sequence of N tokens $w_1, w_2, \dots, w_N$:

$$PP = \exp \left(-\frac{1}{N} \sum_{t=1}^N \log P(w_t | w_{<t}) \right) \quad (22)$$

4.2 Intuitive Interpretation

Perplexity can be interpreted as:

- The average branching factor - how many choices the model effectively has at each step
- The effective vocabulary size the model is choosing from at each position
- A measure of uncertainty: lower perplexity = more confident predictions

4.3 Detailed Example with Small Vocabulary

Let's work through a concrete example with vocabulary $V = \{\text{cat}, \text{dog}, \text{bird}, \text{fish}\}$.

4.3.1 Setup

Consider the sentence: "cat dog bird"

We want to evaluate how well our language model predicts this sequence.

True next-word probabilities (what actually happened):

- After "cat": next word is "dog" (probability = 1.0 for "dog", 0.0 for others)
- After "cat dog": next word is "bird" (probability = 1.0 for "bird", 0.0 for others)

Model's predicted probabilities:

After "cat", model predicts:

Next Word	Predicted Probability
cat	0.1
dog	0.6
bird	0.2
fish	0.1

After "cat dog", model predicts:

Next Word	Predicted Probability
cat	0.2
dog	0.1
bird	0.5
fish	0.2

4.3.2 Calculating Cross-Entropy

The cross-entropy is:

$$CE = -\frac{1}{2}[\log P(\text{dog}|\text{cat}) + \log P(\text{bird}|\text{cat}, \text{dog})] \quad (23)$$

$$= -\frac{1}{2}[\log(0.6) + \log(0.5)] \quad (24)$$

$$= -\frac{1}{2}[-0.511 + (-0.693)] \quad (25)$$

$$= -\frac{1}{2}[-1.204] \quad (26)$$

$$= 0.602 \quad (27)$$

4.3.3 Calculating Perplexity

$$PP = \exp(CE) \tag{28}$$

$$= \exp(0.602) \tag{29}$$

$$= 1.83 \tag{30}$$

4.3.4 Interpretation

A perplexity of 1.83 means:

- On average, the model is as uncertain as if it were choosing uniformly among 1.83 equally likely words
- This is quite good performance - the model is relatively confident in its predictions
- Perfect prediction would give $PP = 1$
- Random guessing among 4 words would give $PP = 4$

4.4 Comparison with Different Model Performance

4.4.1 Better Model

Suppose a better model gives probabilities:

- $P(\text{dog}|\text{cat}) = 0.8$
- $P(\text{bird}|\text{cat}, \text{dog}) = 0.7$

Then:

$$CE = -\frac{1}{2}[\log(0.8) + \log(0.7)] = 0.235 \tag{31}$$

$$PP = \exp(0.235) = 1.26 \tag{32}$$

4.4.2 Worse Model

A worse model with probabilities:

- $P(\text{dog}|\text{cat}) = 0.3$
- $P(\text{bird}|\text{cat}, \text{dog}) = 0.2$

Gives:

$$CE = -\frac{1}{2}[\log(0.3) + \log(0.2)] = 1.105 \tag{33}$$

$$PP = \exp(1.105) = 3.02 \tag{34}$$

4.4.3 Random Model

A model that assigns equal probability to all words:

- $P(\text{dog}|\text{cat}) = 0.25$
- $P(\text{bird}|\text{cat}, \text{dog}) = 0.25$

Gives:

$$CE = -\frac{1}{2}[\log(0.25) + \log(0.25)] = 1.386 \quad (35)$$

$$PP = \exp(1.386) = 4.0 \quad (36)$$

This matches our intuition: random guessing among 4 equally likely words gives perplexity = 4.

5 Practical Considerations

5.1 Perplexity Ranges

- $PP = 1$: Perfect prediction
- $PP \in [10, 20]$: Good language model (typical for modern LLMs)
- $PP \in [20, 50]$: Reasonable performance
- $PP > 100$: Poor performance
- $PP = |V|$: Random performance (where $|V|$ is vocabulary size)

5.2 Relationship to Bits per Token

5.2.1 Derivation of Equation (22)

The relationship between perplexity and bits per token comes directly from information theory. Let's derive this step by step.

Recall that perplexity is defined as:

$$PP = \exp\left(-\frac{1}{N} \sum_{t=1}^N \log P(w_t|w_{<t})\right) \quad (37)$$

Taking the natural logarithm of both sides:

$$\ln(PP) = -\frac{1}{N} \sum_{t=1}^N \log P(w_t|w_{<t}) \quad (38)$$

This is exactly the cross-entropy! So:

$$CE = \ln(PP) \quad (39)$$

Now, in information theory, the amount of information (in bits) required to encode an event with probability p is $-\log_2(p)$. The cross-entropy in base 2 gives us the average bits per token:

$$CE_{\text{base } 2} = -\frac{1}{N} \sum_{t=1}^N \log_2 P(w_t | w_{<t}) \quad (40)$$

Since $\log_2(x) = \frac{\ln(x)}{\ln(2)}$, we have:

$$CE_{\text{base } 2} = \frac{CE}{\ln(2)} = \frac{\ln(PP)}{\ln(2)} \quad (41)$$

Using the change of base formula $\log_2(PP) = \frac{\ln(PP)}{\ln(2)}$:

$$\boxed{\text{Bits per token} = \log_2(PP)} \quad (42)$$

5.2.2 Intuitive Explanation

This relationship tells us:

- If $PP = 2$, we need $\log_2(2) = 1$ bit per token on average
- If $PP = 4$, we need $\log_2(4) = 2$ bits per token on average
- If $PP = 16$, we need $\log_2(16) = 4$ bits per token on average

This makes intuitive sense: if the model is equally uncertain among k choices at each step, it needs $\log_2(k)$ bits to encode each choice.

5.2.3 Worked Example

Using our previous example where $PP = 1.83$:

$$\text{Bits per token} = \log_2(1.83) \quad (43)$$

$$= \frac{\ln(1.83)}{\ln(2)} \quad (44)$$

$$= \frac{0.602}{0.693} \quad (45)$$

$$= 0.87 \text{ bits per token} \quad (46)$$

This means our model needs less than 1 bit per token to encode the sequence, indicating good compression performance.

5.2.4 Comparison with Fixed-Length Encoding

For our 4-word vocabulary, a fixed-length binary encoding would require:

- cat = 00 (2 bits)
- dog = 01 (2 bits)
- bird = 10 (2 bits)
- fish = 11 (2 bits)

So fixed-length encoding uses exactly 2 bits per token. Our model's 0.87 bits per token represents significant compression due to better probability modeling.

5.2.5 Connection to Data Compression

This relationship explains why language models are excellent data compressors:

- Lower perplexity $\Rightarrow$ fewer bits needed per token
- Better probability modeling $\Rightarrow$ better compression
- The cross-entropy lower bound: you cannot compress below $H(P)$ bits per token for true distribution P

5.3 Limitations

- Perplexity doesn't capture semantic quality
- Lower perplexity doesn't always mean better text generation
- Domain-dependent: models achieve different perplexities on different text types

6 Conclusion

Perplexity provides a principled way to evaluate language models by measuring how surprised the model is by the actual text. It builds naturally from information theory concepts:

- **Entropy** measures uncertainty in a distribution
- **Cross-entropy** measures the cost of using wrong probabilities
- **Perplexity** exponentiates cross-entropy to give an interpretable "effective vocabulary size"

Understanding these connections helps us better interpret model performance and make informed decisions about model selection and improvement.